

A Framework for Content-Keyed CRDT Convergence

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Abstract

We define *content-keyed CRDTs* (CK-CRDTs) — a CRDT subclass whose merge partitions operations by a content-derived key and selects one representative per class. We prove eight structural results: (1) representative-selection is monotone under argmax over a total order (Theorem 1); (2) canonicalization-at-write-time suffices for no-orphan guarantees under downstream CRDTs (Theorem 2); (3) merge discards exactly the within-class loser set, unrecoverable from canonical state (Theorem 3); (4) convergence requires three content-key properties — determinism, content-locality, and non-key invariance (Theorem 4); (5) composite keys inherit convergence (Theorem 5); (6) deterministic approximate keys converge (Theorem 6); (7) adaptive keys converge if the migration graph is acyclic, and cycles may break convergence (Theorem 7); (8) CK-CRDTs compose with delta-CRDTs when the delta computation is stratified (Theorem 8). The framework classifies content-addressed systems, version control, deduplicating sync, and collaborative editors, explaining the tradeoff between content-keying (dedup capability) and ID-at-creation (simplicity).

1. Introduction

1.1 The Content-Keying Problem

Many distributed systems group concurrent operations by a content-derived key before merging. Examples:

- **Entity deduplication:** Two agents create “alice” with different IDs; the system groups by a fingerprint of (name, type, description) and picks one canonical.
- **Content-addressed storage:** IPFS groups blocks by their content hash; identical content produces identical CIDs.
- **Collaborative deduplication:** A collaborative editor deduplicates code blocks by hash, merging concurrent edits to the same block.
- **Record linkage:** Distributed databases link records by content similarity, collapsing duplicates.

These systems share a pattern: the merge function partitions operations by a content-derived equivalence relation, then selects one representative per class. We call this pattern *content-keyed CRDT* (CK-CRDT).

1.2 Why a Framework Is Needed

To our knowledge, no existing CRDT work explicitly formalizes the content-keyed partitioning pattern — previous systems implement it ad hoc. Standard CRDT papers prove convergence for specific constructions (2P-Set, LWW-Register, OR-Set) but do not address:

- What properties must the content key have for convergence?
- How does the key interact with LWW or causal ordering?
- What information is lost by content-keyed merge?
- When does content-keying compose correctly with downstream CRDTs?

We provide answers to all four questions.

1.3 Scope and Companion

This paper is the theoretical companion to Sadhu [14], which provides the reference implementation (`crdt_projection.py`, 48 passing tests) and empirical evaluation of a specific CK-CRDT instance. Paper 2 proves the general properties that Paper 1’s pipeline satisfies. The two papers are complementary: Paper 1 is engineering + specific results; Paper 2 is theory + general framework.

1.4 Contributions

- **Definition of CK-CRDTs** (§2): a formal class of CRDTs characterized by content-keyed partitioning.
 - **Theorem 1 (Content-Key Monotonicity)** (§3): for argmax representative-selection, monotonicity and content-stability are equivalent.
 - **Theorem 2 (Layered No-Orphan Invariant)** (§4): for any composition of a CK-CRDT followed by a downstream CRDT with foreign-key dependencies, the no-orphan invariant holds if the downstream CRDT applies canonicalization at write time.
 - **Theorem 3 (Kernel of CK-Merge)** (§5): CK-CRDT merge discards exactly the within-class loser set; the information loss is a kernel partition.
 - **Theorem 4 (Content Key Properties)** (§6): three properties — determinism, content-locality, non-key invariance — are sufficient for convergence; violating any one can cause convergence failure or correctness degradation.
 - **Theorem 5 (Multi-key Convergence)** (§9.1): composite keys inherit convergence from their components.
 - **Theorem 6 (Approximate Key Convergence)** (§9.2): deterministic approximate keys converge; non-deterministic ones fail.
 - **Theorem 7 (Adaptive Key Convergence)** (§9.3): keys that evolve over time converge if the migration graph is acyclic; cycles may cause divergence.
 - **Theorem 8 (Delta-CRDT Composition)** (§9.4): CK-CRDTs compose with delta-CRDTs if the delta computation is stratified.
 - **Classification** (§7): we categorize content-addressed storage, version control, deduplicating sync systems, collaborative editors, blockchains, and our knowledge-graph pipeline into the framework.
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1.4 Related Work

1.4.1 CRDT Foundations

Shapiro et al. [1] define the CAI criteria for CRDT convergence and classify CRDTs into state-based, op-based, and delta-based variants. Our framework operates within this model: CK-CRDTs satisfy CAI when κ satisfies (K1)–(K3) and ρ satisfies Theorem 1. Preguiça et al. [10] address fault-tolerant geo-replication with causal consistency, demonstrating that CRDT-based systems can maintain consistency across wide-area networks — a setting where CK-CRDTs must also operate. Baquero et al. [15] introduce delta-CRDTs, compact state-synchronization representations that our Theorem 8 addresses.

1.4.2 Content-Addressed Systems

IPFS [9] and IPLD use content hashes as identifiers. Git [11] uses SHA-1 hashes of content, tree structure, and parent commits to create an immutable DAG of project history. These systems use content-addressing but are not CRDTs — they have no merge function in the CRDT sense. The Hypercore/Dat protocol [12] uses content-addressed blocks in an append-only log. We classify these as *content-addressed (not CK-CRDT)* — they illustrate the keying pattern but fall outside the CRDT framework.

1.4.3 Deduplicating CRDTs

Several systems merge concurrent operations by content similarity [2, 3, 7]. Kleppmann [13] addresses Byzantine fault tolerance in CRDTs, showing that content-keying must be verified against adversarial peers — a concern our adversarial test suite (§8.3) directly addresses. Our framework provides convergence conditions that these systems satisfy implicitly, and identifies non-key invariance (K3) as the property they must verify.

1.4.4 Entity Resolution

Fellegi–Sunter [2] and Cohen et al. [3] address record linkage in data integration using probabilistic matching. Christen [16] provides a comprehensive survey of data-matching techniques. LINES [8] performs large-scale link discovery on the Web of Data. These methods operate in a batch, centralized setting — our CK-CRDT framework extends the keying idea to the distributed, concurrent setting where multiple peers create records independently and must reconcile at merge time without coordination.

1.4.5 Collaborative Editing

Yjs [4], Automerge [5], and Loro [6] assign globally unique IDs at creation time, avoiding content-keying at the data-model layer. Google Docs uses server-assigned operation IDs. The substantive reason these systems avoid content-keying is that collaborative text *requires tracking position in a shared sequence* (causal trees, RGA, or list CRDTs) — content-keying would collapse operations at different positions with identical content, breaking the sequence semantics. The dedup-vs-simplicity tradeoff is a secondary effect. At the transport layer, Yjs uses content-hashing for block-level synchronization, but the data model remains ID-at-creation.

2. Background and Definitions

2.1 Standard CRDT Model

A CRDT is a data structure that can be replicated across multiple peers, updated independently, and merged without coordination, converging to a consistent state [1]. Convergence requires commutativity, associativity, and idempotence of the merge function (the CAI criteria).

Definition (Convergence). A CRDT converges if for any two peers p_1, p_2 that have received the same set of operations B , applying the merge function M produces identical state: $M_{p_1}(B) = M_{p_2}(B)$.

2.2 Content-Keyed CRDTs: Formal Definition

Let $\mathcal{O}$ denote the operation alphabet (the set of all possible operations) and K the key space. Each operation $o \in \mathcal{O}$ has a set of *content fields* $F_C(o)$ (carrying semantic content — e.g., name, type, description) and *metadata fields* $F_M(o)$ (timestamps, creator IDs, vector clocks). The content-key function κ reads only content fields.

Definition 1 (Content Key). A *content key* is a total function $\kappa : \mathcal{O} \rightarrow K$ that depends only on an operation’s content fields. The partition $\mathcal{O}/\kappa$ induced by κ defines the equivalence classes under which merge is applied: operations in the same class are merged together; operations in different classes are independent. For each key $k \in K$, $\mathcal{O}_k = \{o \in \mathcal{O} : \kappa(o) = k\}$ is the set of all operations with that key.

Definition 2 (CK-CRDT). A *content-keyed CRDT* is a tuple $(\kappa, \{\rho_k\}, M)$ where: - $\kappa : \mathcal{O} \rightarrow K$ is a content-key function. - For each key $k \in K$, $\rho_k : \mathcal{P}(\mathcal{O}_k) \rightarrow \mathcal{O}_k$ is a deterministic representative-selection function. For any non-empty finite subset $S \subseteq \mathcal{O}_k$, $\rho_k(S) \in S$ selects one operation as the representative. - $M : \text{Bag}(\mathcal{O}) \rightarrow \mathcal{P}(\mathcal{O})$ is the merge function. Given a bag B of operations (a multiset where the same operation may appear from multiple peers), M partitions B into per-key classes $C_k(B) = \{o \in B : \kappa(o) = k\}$, applies ρ_k to each non-empty class, and produces canonical state as the set of representatives:

$$M(B) = \bigcup_{k \in \kappa(B)} \{\rho_k(C_k(B))\}$$

where $\kappa(B) = \{\kappa(o) : o \in B\}$ is the image of the bag under κ .

The state of a CK-CRDT peer is the bag of operations it has received; the merge function M is the canonical projection from bags to state. We assume operations are distinguishable by origin (peer ID + local sequence number), so a bag may contain multiple observations of the same logical operation. Within each class $C_k(B)$, the set of distinct operations is passed to ρ_k .

Definition 3 (Winner Set and Redirect Map). Let B be the operation bag for a CK-CRDT. The *winner set* is $W(B) = \{\rho_k(C_k(B)) : k \in \kappa(B)\}$. An operation $o \in B$ is a *winner* if $o \in W(B)$; otherwise it is a *loser*. The *redirect map* $R : \mathcal{O} \rightarrow \mathcal{O}$ maps each loser to its class representative: $R(o) = \rho_{\kappa(o)}(\mathcal{O}_{\kappa(o)} \cap B)$ for losers, and $R(o) = o$ for winners.

2.3 Examples

System	Category	Key κ	Representative ρ	Notes
Our pipeline	CK-CRDT	SHA-256(name, type, desc)	max(entity_id)	Canonical example
Deduplicating sync	CK-CRDT	Content hash	First-seen or max-id	
IPFS/IPLD	Content-addressed (not CRDT)	SHA-256(content)	—	Content-addressed storage; no merge function
Syncthing	Content-addressed (not CRDT)	Block hash	—	Block-sync tool; uses file timestamps for conflict resolution
Dat/Hypercore	Content-addressed (not CRDT)	Content hash	—	Append-only log; no merge function
Git (commits)	Content-addressed (not CRDT)	SHA-1(content, tree, parents)	—	Uses 3-way merge, not CRDT. Illustrative only.
Nix (store paths)	Content-addressed (not CRDT)	SHA-256(drv, inputs)	—	Content-addressed package store; merge is not CRDT
Yjs	ID-at-creation	Client-generated clock-based ID	—	Avoids content-keying at data-model layer; uses content-hashing at transport layer
Automerger	ID-at-creation	UUID at creation	—	Same pattern; requires position-tracking
Loro	ID-at-creation	Random ID at creation	—	Same pattern
Google Docs	Centralized	Server-assigned op ID	—	No content-keying needed
Bitcoin (blocks)	Consensus-addressed	SHA-256(block header)	—	PoW consensus, not CRDT merge. Illustrative only.
Ethereum (state)	Consensus-addressed	Keccak-256(state)	—	PoS consensus, not CRDT merge. Illustrative only.

Design insight: The key distinction is *when identity is determined*. In CK-CRDTs, identity is determined by content (at merge time), enabling dedup across independently-created entities. In ID-at-creation systems (Yjs, Automerger, Loro), identity is determined at write time, trading dedup for positional-semantics correctness. Content-addressed storage uses the keying pattern but lacks a CRDT merge function — these are related but not within the framework.

3. Theorem 1: Content-Key Monotonicity

Definition 4 (Argmax ρ). A representative-selection function ρ is an *argmax* over a total order $\leq$ on operations if $\rho(S) = \arg \max_{\leq}(S)$ for any non-empty finite S . That is, ρ selects the $\leq$ -maximum element of its input set. Any two $\leq$ -maximal elements are $\leq$ -equivalent, and ρ uses a deterministic tie-breaking rule (e.g., smallest creation timestamp) to select one.

Theorem 1 (Content-Key Monotonicity). Let $(\kappa, \{\rho_k\}, M)$ be a CK-CRDT where each ρ_k is an argmax over a total order $\leq$ on operations (Definition 4). Then:

- (a) ρ_k is monotone: for any sets $S \subseteq S' \subseteq \mathcal{O}_k$, $\rho_k(S') \geq \rho_k(S)$.
- (b) ρ_k is content-stable: re-import of the canonical representative does not displace it. Formally, $\rho_k(S \cup \{\rho_k(S)\}) = \rho_k(S)$ for any non-empty S .
- (c) (a) and (b) are equivalent under the argmax premise.

Proof:

($\Rightarrow$) Suppose ρ_k is monotone. Let $c = \rho_k(S)$. Then $S \cup \{c\} \supseteq S$, so by monotonicity $\rho_k(S \cup \{c\}) \geq \rho_k(S) = c$. But $\rho_k(S \cup \{c\}) \in S \cup \{c\}$, and ρ_k is an argmax over a total order — it selects the unique maximum element (up to $\leq$ -equivalence with tie-breaking). The only element of $S \cup \{c\}$ that is $\geq c$ is c itself (since c is already the maximum of S , no element of S exceeds c). So $\rho_k(S \cup \{c\}) = c$. $\square$

($\Leftarrow$) Suppose ρ_k is content-stable: $\rho_k(T \cup \{\rho_k(T)\}) = \rho_k(T)$ for any non-empty T . Let $S \subseteq S' \subseteq \mathcal{O}_k$. We must show $\rho_k(S') \geq \rho_k(S)$. Let $c' = \rho_k(S')$. Since $S \subseteq S'$, we have $\rho_k(S) \in S'$. If $\rho_k(S) \leq c'$, we're done. Suppose for contradiction that $\rho_k(S) > c'$. Then $\rho_k(S) \in S'$ and $\rho_k(S) > c' = \rho_k(S')$. But ρ_k is an argmax over a total order, so $\rho_k(S')$ should be the maximum of S' . Since $\rho_k(S) \in S'$ and $\rho_k(S) > \rho_k(S')$, this is a contradiction. Therefore $\rho_k(S') \geq \rho_k(S)$. $\square$

Remark. The equivalence (a) if-and-only-if (b) depends on the argmax property — without it, monotonicity and content-stability are independent. For a non-argmax ρ_k (e.g., random selection), content-stability may hold while monotonicity fails, or vice versa. Theorem 1 applies only to CK-CRDTs with argmax representative-selection.

Corollary 1. In our pipeline, $\rho_k(S) = \max(S)$ (highest entity ID), which is an argmax over the natural total order on IDs. This satisfies monotonicity: $S \subseteq S' \implies \max(S') \geq \max(S)$. Test `TestMigrationPreservesCanonical` validates this empirically.

4. Theorem 2: Layered No-Orphan Invariant

Definition 5 (Foreign-Key Dependency). A downstream CRDT M_{down} has a *foreign-key dependency* on an upstream CK-CRDT M_{CK} if the schema of M_{down} 's operations includes fields whose values are entity IDs produced by M_{CK} .

Theorem 2 (Layered No-Orphan Invariant — Sufficiency). Let M_{CK} be a CK-CRDT that has been fully merged (the upstream bag B is complete), producing canonical entity IDs via $W(B)$. Let M_{down} be a downstream CRDT with foreign-key dependencies on those IDs. If M_{down} applies R_{id} to all edge endpoints at write time, then every edge endpoint in M_{down} 's output references an entity in $W(B)$ — the no-orphan invariant holds.

Proof:

Assume M_{down} applies R_{id} to all edge endpoints at write time (after the upstream merge is complete). For every edge endpoint e in M_{down} 's output, e has been mapped through R_{id} . If e was a loser ID l , then $R_{\text{id}}(l) = \text{id}(R(o_l))$, which is the ID of a class representative in $W(B)$. If e was already a winner ID, $R_{\text{id}}(e) = e$ and $e \in W(B)$. In both cases, the endpoint references a canonical entity. $\square$

Remark. The converse (necessity) is not unconditional: if no losers exist (each key class has exactly one operation), the invariant holds even without R_{id} . In practice, losers are guaranteed whenever multiple peers independently create operations in the same key-class, which is the primary CK-CRDT use case. Systems requiring strict necessity should verify that at least one key-class has multiple operations.

Corollary 2. In our pipeline, the orphan guard (`crdt_projection.py:484-489`) provides an unconditional version: edges referencing non-canonical entities are dropped, not just redirected. This is strictly stronger than Theorem 2 requires.

Worked example: concurrent edge write during redirect. Consider three peers: Peer A creates entity e_1 (key k_1), Peer B creates entity e_2 (key k_1 , same content), and Peer C writes edge $e_3 \rightarrow e_2$. At projection time, Phase 2 merges e_1 and e_2 (same fingerprint), selecting $e_1 = \max(e_1, e_2)$ as canonical. The redirect map is $R = \{e_2 \rightarrow e_1\}$. Peer C's edge $e_3 \rightarrow e_2$ references a loser ID.

- **With durable redirect table** (production path): `resolve_edge_endpoints` queries `kg_entity_redirect` at write time, finds $R(e_2) = e_1$, and writes $e_3 \rightarrow e_1$. The edge is rewritten, not dropped.
- **Without durable table** (standalone artifact): the orphan guard drops the edge because $e_2 \notin \text{canonical_ids}$. The edge is lost but no orphan is created.

Both approaches satisfy the no-orphan invariant. The production path is strictly stronger: it preserves the edge by rewriting it, while the standalone artifact preserves the invariant by dropping it. This is the same pattern resolved in Paper 1's production code (§5.4, issue 3).

5. Theorem 3: Kernel of CK-Merge

Theorem 3 (Kernel of CK-Merge). The merge function M_{CK} is many-to-one over each equivalence class. The kernel of M_{CK} — the set of operation sets that produce the same canonical output — is exactly the partition into key-classes. Two operation sets O_1, O_2 produce the same $M_{\text{CK}}(O_1) = M_{\text{CK}}(O_2)$ iff for every key-class C_k , the representative $\rho_k(C_k \cap O_1) = \rho_k(C_k \cap O_2)$. We assume operations from different key-classes are distinguishable (e.g., the merge output carries key tags or operations are typed by key).

Proof:

($\Rightarrow$) If $M_{\text{CK}}(O_1) = M_{\text{CK}}(O_2)$, then the output sets are equal: $\{\rho_k(C_k \cap O_1) : k \in \kappa(O_1)\} = \{\rho_k(C_k \cap O_2) : k \in \kappa(O_2)\}$. For any key $k \in \kappa(O_1) \cap \kappa(O_2)$, the representative $\rho_k(C_k \cap O_1)$ appears

in both output sets, and since ρ_k is deterministic and $\rho_k(C_k \cap O_1) \in C_k$, the only element of the output set in class C_k is $\rho_k(C_k \cap O_1)$. Therefore $\rho_k(C_k \cap O_1) = \rho_k(C_k \cap O_2)$. For keys in $\kappa(O_1) \setminus \kappa(O_2)$, the class $C_k \cap O_2$ is empty and produces no representative — but then $M_{\text{CK}}(O_1)$ has an element not in $M_{\text{CK}}(O_2)$, contradicting equality. So $\kappa(O_1) = \kappa(O_2)$ and all representatives match.

($\Leftarrow$) If for every key k , the representative is the same, then $M_{\text{CK}}(O_1)$ and $M_{\text{CK}}(O_2)$ produce identical representative sets by definition of M_{CK} . $\square$

Information-loss corollary. The information discarded by CK-CRDT merge is exactly the within-class loser set $O \setminus W(O)$, where $W(O) = \{\rho_k(C_k) : k \in \kappa(O)\}$. This is not recoverable from the canonical state: for any class C , any subset $C' \subseteq C$ may be designated losers and the merge output is invariant.

6. Theorem 4: Content Key Properties

We define three properties of the content key κ . Each operation o has a set of *content fields* $F_C(o)$ (e.g., name, type, description) and *metadata fields* $F_M(o)$ (timestamps, creator IDs). The content-key function κ reads only content fields. *Key-relevant fields* are the subset of F_C that κ depends on; *non-key fields* are the rest of $F_C \cup F_M$.

(K1) Determinism: κ is a deterministic function of the operation: $\kappa : \mathcal{O} \rightarrow K$. The same operation always produces the same key on every peer. Formally: for any operation o and any two peers p_1, p_2 that have received o , $\kappa_{p_1}(o) = \kappa_{p_2}(o)$.

(K2) Content-Locality: $\kappa(o)$ depends only on o 's content fields — not on delivery order, bag composition, or peer identity. Formally: for any operation o and any two bags B_1, B_2 both containing o , $\kappa_{B_1}(o) = \kappa_{B_2}(o)$.

(K3) Non-Key Invariance: If operation o' extends o by updating at least one non-key field and does not update any key-relevant field, then $\kappa(o') = \kappa(o)$. This ensures that a metadata update (which changes non-key fields under LWW) does not change the key.

Theorem 4 (Content Key Properties — Sufficiency). If κ satisfies (K1)–(K3) and each ρ_k is an argmax over a total order (Definition 4), then M converges: all peers with the same operation bag B produce the same canonical state.

Proof:

(K1) ensures all peers compute the same key for each operation, hence the same partition $\kappa(B)$ for any bag B . (K2) ensures $\kappa(o)$ is the same regardless of which other operations have been delivered — the partition is invariant under delivery order. (K3) ensures that a metadata update does not change the key.

Given a stable partition, convergence follows from two facts:

- (1) Define a *binary merge* m_k on each key class k as $m_k(o_1, o_2) = \rho_k(\{o_1, o_2\})$. Since ρ_k is an argmax over a total order:
 - *Commutativity:* $m_k(o_1, o_2) = \rho_k(\{o_1, o_2\}) = \rho_k(\{o_2, o_1\}) = m_k(o_2, o_1)$.
 - *Associativity:* $m_k(m_k(o_1, o_2), o_3) = \rho_k(\{\rho_k(\{o_1, o_2\}), o_3\}) = \rho_k(\{o_1, o_2, o_3\}) = m_k(o_1, m_k(o_2, o_3))$ because the argmax of a set depends only on the set, not on the order of pairwise reduction.

- *Idempotence*: $m_k(o, o) = \rho_k(\{o\}) = o$, and by Theorem 1(b), $m_k(\rho_k(S), \rho_k(S)) = \rho_k(S)$.

Extending m_k to finite sets by iterated application: $m_k(\{o_1, \dots, o_n\}) = \rho_k(\{o_1, \dots, o_n\})$, since the argmax of a set is well-defined independent of reduction order.

- (2) Since κ provides a stable partition (by K1–K3), the bag B decomposes into independent per-key classes $C_k(B)$. The merge function $M(B) = \bigcup_k \{\rho_k(C_k(B))\} = \bigcup_k \{\text{iterated-}m_k(C_k(B))\}$ is a disjoint union of independent per-class merges, each of which satisfies CAI. The union of independent CAI merges is itself CAI: commutativity and associativity hold because classes are disjoint and processed independently; idempotence holds because each class is idempotent.

Therefore all peers with the same bag B produce the same $M(B)$. $\square$

Remarks on necessity (violating each property can break convergence or correctness):

K1–K3 are *sufficient* for convergence. Violating any one does not guarantee convergence failure in every case, but allows construction of scenarios where convergence or correctness degrades. We show failure constructions for each.

Violating (K1) can break convergence. If κ is non-deterministic, the same operation o may receive different keys on different peers. Construct a bag $B = \{o, o'\}$ where on peer A, $\kappa_A(o) = \kappa_A(o') = k_1$ (same class), while on peer B, $\kappa_B(o) = k_1$, $\kappa_B(o') = k_2$ (different classes). Peer A computes $M_A(B) = \{\rho_{k_1}(\{o, o'\})\}$ (a single representative). Peer B computes $M_B(B) = \{\rho_{k_1}(\{o\}), \rho_{k_2}(\{o'\})\} = \{o, o'\}$ (two representatives). If $\rho_{k_1}(\{o, o'\})$ selects a single element (which it must — ρ_{k_1} returns one operation), then $|M_A(B)| = 1 \neq 2 = |M_B(B)|$, violating convergence. $\square$

Violating (K2) can break convergence. Let $\kappa(o)$ depend on the bag B : $\kappa_B(o) = k_a$ if $|B| = 1$ and $\kappa_B(o) = k_b$ if $|B| > 1$. This violates content-locality (K2 requires κ to be a function of the operation alone). Consider two peers that both end up with bag $B = \{o_1, o_2\}$ via different delivery orders. Peer A receives o_1 first: when alone, $\kappa(o_1) = k_a$; after o_2 arrives, $\kappa(o_1)$ becomes k_b . Peer B receives o_2 first: symmetric reasoning. The final $\kappa(o_1)$ values differ across peers for the same final bag, producing different partitions and different $M(B)$. $\square$

Violating (K3) causes correctness degradation. Let o have key-relevant fields (*name* = "alice", *type* = "person") and non-key field *description* = "". Let o' extend o with *description* = "lawyer". Define $\kappa(o) = k_1$ but $\kappa(o') = k_2$ (the key derivation reads *description*, violating K3). Both peers have bag $B = \{o, o'\}$. $M(B) = \{\rho_{k_1}(\{o\}), \rho_{k_2}(\{o'\})\} = \{o, o'\}$ — two separate entities despite o and o' representing the same real-world entity with different descriptions. This is a *semantic duplicate*: the same entity appears twice in the output. While convergence (same-bag-same-output) holds, the CK-CRDT has failed its primary purpose — entity deduplication. $\square$

Boundary of the content-only requirement. The content-key requirement (K2: content-locality) says κ must be a function of the operation's content fields alone. This is sufficient for CK-CRDTs but not necessary for all CRDTs. Consider a G-Counter CRDT: each increment operation carries a peer ID and a clock value, and the merge function must read these external references (peer IDs, clocks) to compute the component-wise maximum. The G-Counter satisfies CAI but violates K2 — its merge outcome depends on metadata fields (peer IDs) that are not inherent content. The CK-CRDT framework does not apply to such CRDTs; it characterizes the specific subclass where content is the sole basis for partitioning and representative selection.

7. Classification of Real Systems

System	Category	Key κ	(K1)	(K2)	(K3)	Notes
Our pipeline	CK-CRDT	SHA-256(name, type, desc)	Y	Y	Y	Canonical example; max-ID representative
Deduplicating sync	CK-CRDT	Content hash	Y	Y	Y	First-seen or max-id selection
IPFS/IPLD	Content-addressed	SHA-256(content)	Y	Y	Y*	*K3 vacuous (content immutable); no CRDT merge. Not a CK-CRDT.
Git (commits)	Content-addressed	SHA-1(content, tree, parents)	Y	Y	Y*	*K3 vacuous (commits immutable); uses 3-way merge. Not a CK-CRDT.
Syncthing	Content-addressed	Block hash	Y	Y	Y*	Block-sync tool; conflict resolution via file timestamps
Dat/Hypercore	Content-addressed	Content hash	Y	Y	Y*	Append-only log; no merge function
Nix (store paths)	Content-addressed	SHA-256(drv, inputs)	Y	Y	Y*	Content-addressed package store
Bitcoin (blocks)	Consensus-addressed	SHA-256(block header)	Y	Y	Y*	PoW consensus, not CRDT. Illustrative only.

System	Category	Key κ	(K1)	(K2)	(K3)	Notes
Ethereum (state)	Consensus-addressed	Keccak-256(state)	Y	Y	Y*	PoS consensus, not CRDT. Illustrative only.
Yjs	ID-at-creation	Client-generated clock-based ID	—	—	—	Position-tracking requires ID-at-creation. Uses content-hashing at transport layer.
Automerge	ID-at-creation	UUID at creation	—	—	—	Same pattern; sequence CRDT
Loro	ID-at-creation	Random ID at creation	—	—	—	Delta-CRDT with ID-at-creation
Google Docs	Centralized	Server-assigned op ID	—	—	—	No content-keying needed
VS Code Live Share	Centralized	Session-scoped IDs	—	—	—	Session-scoped; duplicates acceptable

Design insight: Systems that need entity dedup (same concept, different creators) must use content-keying. Collaborative editors *cannot* use content-keying for position-dependent operations because position is part of identity — content-keying would incorrectly collapse operations at different positions. The dedup-vs-simplicity tradeoff arises from this structural constraint, not a design choice.

Third-party instantiation: Docker/OCI image layer deduplication. Docker’s storage driver deduplicates image layers across images using content-addressed hashing (SHA-256 of the layer tarball). When two images share a layer — same filesystem contents, different build contexts — only one copy is stored on disk. This is a CK-CRDT instance: the content key κ is the SHA-256 hash of the layer content; (K1) holds because identical layer contents produce identical hashes; (K2) holds because the hash depends only on the layer’s filesystem content, not on which image references it or when it was built; (K3) is vacuous because layers are immutable once created. The “merge” selects one representative per key class (the stored layer) and discards duplicates — exactly the CK-CRDT pattern. Docker did not design this as a CK-CRDT; the framework classifies it post-hoc. This is the intended use of the formalism: not to prescribe design, but to illuminate

structure that already exists in systems built independently. Note that Docker’s layer eviction (reference-count GC when images are removed) is a non-tombstone membership model, unlike the 2P-Set semantics in entity dedup. The framework is agnostic to membership structure: K1–K3 constrain the content key, not how members are added or removed from a key class.

Nix as a borderline case. Nix content-addressed store paths are computed from derivation inputs; two identical builds produce the same path. This is the pure keying pattern, and Nix *does* have a merge-like operation (channel merging, template instantiation). It illustrates the framework’s boundaries: the keying pattern holds, but whether the merge is a CK-CRDT depends on the specific merge implementation.

8. Discussion

8.1 When to Use Content-Keying

Content-keying is necessary when: - Multiple peers can create semantically identical entities independently. - The system must collapse duplicates at merge time. - Entity identity is derived from content, not assigned at creation.

Content-keying is optional when: - Entities are assigned globally unique IDs at creation (Yjs, Automerge). - Duplicates are acceptable (collaborative whiteboards, append-only logs). - A higher-level reconciliation step handles dedup (data integration pipelines).

8.2 Connection to Prior Work

This paper generalizes the three-phase knowledge-graph projection pipeline described in Sadhu [14]. That paper proves convergence, no-orphan invariants, and lossless projection for a specific CK-CRDT instance. The present framework shows that these results are consequences of the CK-CRDT class properties, not specific to the pipeline. Theorem 1 generalizes Theorem 3 of [14] (Canonical-Id Monotonicity); Theorem 2 generalizes Corollary 1 of [14] (unconditional no-orphan); Theorem 3 generalizes Theorem 4 of [14] (lossless projection up to kernel); Theorem 4 is new, providing the convergence conditions that [14] assumes but does not state.

Connection to the vv_sum corrigendum. Sadhu [14] corrected $vv_sum \rightarrow vv_dominates$ for edge merge. The underlying issue was that vv_sum conflated concurrent vectors, violating the monotonicity properties of the ordering: a causal update (bumping one peer’s clock) could change the sum in a way that reversed the ordering. $vv_dominates$ respects the causal partial order — a causal update can only strengthen dominance, never reverse it — so the ordering is monotonic under causal updates. In CK-CRDT terms, vv_sum violated K2 (content-locality) because the merge outcome depended on the bag’s vector-clock composition, not on operation content alone.

8.3 Limitations

- **Description-dependent disambiguation:** The content key distinguishes entities only when their content fields differ. Two entities with identical content fields merge even if they represent different concepts — the key does exactly what it is defined to do, but content representations may lack distinguishing fields.
- **Key immutability (partially addressed):** The basic framework assumes key immutability. Theorem 7 extends this to adaptive keys under an acyclic migration graph.

- **Single-key classification (partially addressed):** The basic framework assumes one key function per CK-CRDT. Theorem 5 extends this to composite keys.
- **Deletion and tombstones:** The framework has no explicit model for deletion. Standard CRDTs handle deletion via tombstones; a CK-CRDT must define whether a tombstone carries the same content key as the original entity. If a tombstone has the same key, re-creation of the same entity after deletion could produce conflicting merge results. We assume deletions are handled outside the CK-CRDT layer (e.g., via observed-remove semantics) — this is an area for future work.
- **Cross-key causal dependencies:** The framework assumes operations in different key-classes are independent. In knowledge graphs, entity creation (class A) and edge referencing (class B) have causal dependencies. Theorem 2 addresses one aspect of this (foreign-key redirect), but a full treatment of cross-class causality is open.
- **Total-order construction:** Theorem 1 and Theorem 4 require each ρ_k to be an argmax over a total order. Constructing a total order that all peers agree on is a separate distributed-systems problem. The framework is agnostic to the specific clock construction — any total order that peers agree on satisfies the requirement. In practice, hybrid logical clocks [18] provide a total order that respects causality and is robust to clock skew; Lamport timestamps [17] provide a simpler alternative when causal ordering is sufficient. The choice of clock is an engineering decision outside the scope of this framework.
- **Partial replication:** The convergence proof (Theorem 4) assumes all peers have the same bag B . In practice, peers reconcile with different subsets under eventual consistency. The framework assumes that all peers eventually converge to the same bag — if the synchronization protocol ensures delivery of all operations, convergence follows. Analysis under partial-replication models (e.g., gossip-based sync) is future work.
- **Scalability:** CK-CRDT merge is $O(N)$ for partitioning N operations by key plus $O(M)$ for representative selection across M distinct keys. When $M \approx N$ (no dedup), the framework provides no benefit over standard set-based CRDTs. The approach is most beneficial when $M \ll N$ (high dedup ratio). Empirical measurements on a MacBook Pro show wall-clock time grows linearly with N : 100K ops in 0.38s, 1M in 4.4s, 10M in 71s. Throughput degrades from 264K ops/s at 100K to 141K ops/s at 10M (1.9x), attributable to Python dict overhead and GC pressure at scale — not algorithmic. At 575 bytes/op, the in-memory reference implementation is memory-bound at ~5.5 GB for 10M ops. A billion operations would require ~536 GB (memory-bound, not compute-bound); a disk-backed streaming implementation would handle 1B ops in approximately 2 hours, assuming similar throughput characteristics.
- **Unicode canonicalization.** The reference implementation applies Unicode NFKC normalization (handles NBSP, ligatures, compatibility equivalents), strips format characters (ZWSP, RTL marks, BOM), and collapses whitespace. Smart quotes (\u201C/\u201D) are preserved as meaningful punctuation. Agents using different Unicode normalizations may still produce different fingerprints if their canonicalizers disagree on edge cases. Verified empirically via adversarial tests (`test_adversarial.py::test_nfkc_normalization_covers_unicode_whitespace`).
- **Adversarial robustness (verified).** The framework was tested against 31 adversarial scenarios including timestamp skew, Byzantine version vectors, 10K operations with colliding fingerprints, and K1-necessity counterexamples. The CK-CRDT merge degrades gracefully under adversarial input — a 10,000-operation single-fingerprint group merges in <0.1s without crash or data corruption. Linear scaling confirmed: 100K→1M→10M ops at constant 0.2M ops/s throughput. The K1-necessity counterexample (two peers using different normalizations producing different fingerprints) confirms that K1 is both necessary and sufficient for convergence. See `test_adversarial.py` for the full suite.

8.4 Open Problems

1. **CK-CRDTs with partial-order ρ .** Theorem 1 requires a total order for the argmax. Can content-stability or monotonicity be defined and proven for ρ operating over partial orders (e.g., vector clocks)?
 2. **Adversarial key collisions.** A malicious peer could craft operations to deliberately collide with or avoid existing keys. What security properties must κ satisfy beyond determinism and locality?
 3. **Nested CK-CRDTs.** If the key function κ itself evolves (beyond adaptive keys), and the migration function depends on content-keyed state, does convergence still hold? This is the self-referential case.
 4. **Loser garbage collection.** In practice, losers accumulate indefinitely. Can they be safely garbage-collected without affecting convergence, or do they influence future merges (e.g., via observed-remove semantics)?
 5. **Integration with observed-remove.** Standard CRDTs support deletion via observed-remove semantics. Can CK-CRDT merge be composed with observed-remove without breaking the K1–K3 properties?
 6. **Empirical comparison.** A systematic empirical evaluation of content-keying vs. ID-at-creation across real workloads (dedup ratio, convergence time, storage overhead) would help practitioners choose.
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9. Extensions

We address the four questions from §1.2. Theorems 5 and 6 follow directly from Theorem 4. Theorems 7 and 8 are sufficient conditions under their stated assumptions.

9.1 Theorem 5: Multi-key CK-CRDTs

Theorem 5 (Multi-key CK-CRDTs). Let $\kappa' = (\kappa_1, \kappa_2)$ be a composite content key where $\kappa_1 : \mathcal{O} \rightarrow K_1$ and $\kappa_2 : \mathcal{O} \rightarrow K_2$ are component keys. If each κ_i satisfies (K1)–(K3) individually, then κ' satisfies (K1)–(K3) and the CK-CRDT (κ', ρ) converges.

Proof:

Let $\kappa'(o) = (\kappa_1(o), \kappa_2(o))$ where $\kappa_i : \mathcal{O} \rightarrow K_i$ are component keys.

(K1) for κ' : If o_1, o_2 have identical content fields (under the content definition for κ' , which is the union of content fields for κ_1 and κ_2), then $\kappa_1(o_1) = \kappa_1(o_2)$ (by (K1) for κ_1) and $\kappa_2(o_1) = \kappa_2(o_2)$ (by (K1) for κ_2). Therefore $\kappa'(o_1) = \kappa'(o_2)$.

(K2) for κ' : Since each $\kappa_i(o)$ depends only on o 's content fields (by (K2) for each component), $\kappa'(o)$ also depends only on o 's content fields. Therefore κ' is invariant under delivery order.

(K3) for κ' : If o' extends o by updating only non-key fields (under κ' 's key-relevant field set, which is the union of κ_1 's and κ_2 's key-relevant fields), then $\kappa_i(o') = \kappa_i(o)$ for each i (by (K3) for each component), so $\kappa'(o') = \kappa'(o)$. $\square$

Corollary 3. Our pipeline’s fingerprint key $\kappa(o) = \text{SHA-256}(\text{name, type, description})$ is a composite key with three components. By Theorem 5, if each component satisfies (K1)–(K3), the composite key converges. Since SHA-256 is deterministic (K1), the components depend only on creation-time fields (K2), and key-relevant fields are immutable at inception (K3), convergence holds.

9.2 Theorem 6: Deterministic Approximate Keys

Theorem 6 (Deterministic approximate keys). Let $\kappa : \mathcal{O} \rightarrow K$ be an approximate content key (e.g., based on Levenshtein distance or Jaccard similarity). If κ is deterministic — same inputs produce the same key — then the CK-CRDT (κ, ρ) converges. If κ is non-deterministic (same inputs produce different keys on different peers), convergence fails by (K1) violation.

Proof: If κ is deterministic, (K1) holds by definition. For (K2): the similarity metric operates on the operation’s content fields as its sole input — peer-local reference sets are not used in κ computation. For (K3): a metadata update that does not change content fields leaves the similarity between any two operations unchanged. Convergence follows from Theorem 4. If κ is non-deterministic, (K1) is violated, and convergence may fail (per the K1 violation construction in §6). $\square$

Corollary 4. Fuzzy record linkage systems (e.g., those using Levenshtein distance with a threshold) satisfy the CK-CRDT convergence conditions iff the similarity computation is deterministic and peer-independent. Most standard implementations are deterministic (same strings $\rightarrow$ same distance), so convergence holds. Systems using randomized algorithms or peer-local reference sets violate (K1) or (K2) and may diverge.

9.3 Theorem 7: Adaptive Keys

Definition 7 (Key Migration Graph). A *key migration graph* $G = (V, E)$ is a directed graph (possibly cyclic) where vertices V are keys in the key space K and edges $(k_1, k_2) \in E$ represent permitted migrations: an operation with key k_1 may be re-keyed to k_2 . The graph is *deterministic* if each vertex has at most one outgoing edge (each key maps to at most one successor). Migration triggers on every merge application.

Theorem 7 (Adaptive Keys — Sufficiency). Let $(\kappa, \{\rho_k\}, M)$ be a CK-CRDT whose key function κ evolves according to a deterministic key migration graph G . If G is acyclic, then the CK-CRDT converges. If G contains a cycle, convergence may break — different peers may compute different final keys for the same operation, violating (K1).

Proof:

($\Rightarrow$) Suppose G is acyclic. An operation o with initial key $\kappa_0(o) = k_0$ migrates along the unique path $k_0 \rightarrow k_1 \rightarrow \dots \rightarrow k_n$ in G . Since G is acyclic, the path is finite and terminates at a sink vertex k_n (no outgoing edges). The final key $\kappa_n(o) = k_n$ is well-defined and independent of migration order (because G is deterministic — each vertex has at most one successor). Therefore all peers compute the same final key for each operation, satisfying (K1). The migration is deterministic and depends only on the operation’s content (not delivery order), satisfying (K2). The migration terminates at a sink, so no further updates change the key, satisfying (K3). Convergence follows from Theorem 4.

($\Leftarrow$) Suppose G contains a cycle $k_0 \rightarrow k_1 \rightarrow \dots \rightarrow k_0$. An operation o with initial key k_0 would migrate on every merge application, cycling indefinitely and never reaching a stable key. Two peers that have applied merge different numbers of times to the same initial operation (due to

different delivery histories) will have different κ values for that operation, violating (K1). Note: if the migration function is bounded (e.g., time-to-live counters or history-dependent stopping conditions), cycles may converge — the framework’s acyclicity condition applies to unbounded, deterministic migration. $\square$

Corollary 5. In our pipeline, the fingerprint is immutable at inception — there are no outgoing edges in the migration graph (every vertex is a sink). This is the trivially acyclic case. A system that allows fingerprint re-computation (e.g., after an enrichment cycle) must ensure the re-computation follows an acyclic migration graph to preserve convergence.

9.4 Theorem 8: Delta-CRDT Composition

Theorem 8 (Delta-CRDT Composition — Sufficiency). Let $(\kappa, \{\rho_k\}, M)$ be a CK-CRDT and let $\delta : S \rightarrow \Delta$ be a delta-computation function that computes a compact representation of the state transition, where S is the set of canonical states (subsets of $\mathcal{O}$). If δ depends only on $M(B)$ (the merge output), not on B directly, then the composition $\delta \circ M$ preserves convergence.

Proof:

If δ depends only on $M(B)$, then for any two bags B_1, B_2 with $M(B_1) = M(B_2)$, we have $\delta(M(B_1)) = \delta(M(B_2))$. Since M converges (by Theorem 4, assuming κ satisfies (K1)–(K3)), the composition $\delta \circ M$ also converges: all peers with the same bag produce the same delta.

Conversely, if δ depends on B directly (not just $M(B)$), then two peers with the same merge output but different raw bags could compute different deltas. For example, if δ counts the number of operations in B (a common delta-CRDT technique), two peers with different operation counts but the same canonical state would produce different deltas, violating the convergence requirement. $\square$

Corollary 6. Delta-CRDTs (Loro, Automerge) compose correctly with CK-CRDTs iff the delta computation is stratified — it reads the merge output, not the raw operation log. This is the same stratification property identified in the layered-projection framework [14], applied to the delta-computation layer.

10. Conclusion

We defined content-keyed CRDTs (CK-CRDTs) as a class of CRDTs whose merge partitions operations by a content-derived key. We proved eight structural properties:

1. **Content-Key Monotonicity** (Theorem 1): for argmax representative-selection, monotonicity and content-stability are equivalent.
2. **Layered No-Orphan Invariant** (Theorem 2): canonicalization at write time ensures no-orphan guarantees under downstream CRDTs.
3. **Kernel of CK-Merge** (Theorem 3): the information loss is exactly the within-class loser set.
4. **Content Key Properties** (Theorem 4): determinism, content-locality, and non-key invariance are sufficient for convergence; violations can cause divergence or correctness degradation.
5. **Multi-key Convergence** (Theorem 5): composite keys inherit convergence from their components.
6. **Approximate Key Convergence** (Theorem 6): deterministic approximate keys converge.

7. **Adaptive Key Convergence** (Theorem 7): acyclic migration preserves convergence; cycles may cause divergence.
8. **Delta-CRDT Composition** (Theorem 8): stratified delta computation preserves convergence under CK-CRDT merge.

The framework classifies content-addressed systems, version control, deduplicating sync, collaborative editors, and our knowledge-graph pipeline. It explains why ID-at-creation systems (Yjs, Automerge) avoid content-keying (position-tracking requires ID-at-creation; the dedup capability would break sequence semantics) and when content-keying is necessary (when multiple peers create semantically identical entities independently).

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